

Complete Design Walkthrough of an EER Diagram for Retail Chain Management System

SUPERMARKET

CSA0508 – DATABASE MANAGEMENT SYSTEM

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DEPARTMENT: AI & ML

COURSE: DATABASE MANAGEMENT SYSTEM

COURSE CODE: CSA0508



What is Retail Chain Management?

Retail Chain Management is a system used to manage multiple retail stores by handling:

- Products
- Customers
- Employees
- Orders

GOAL:

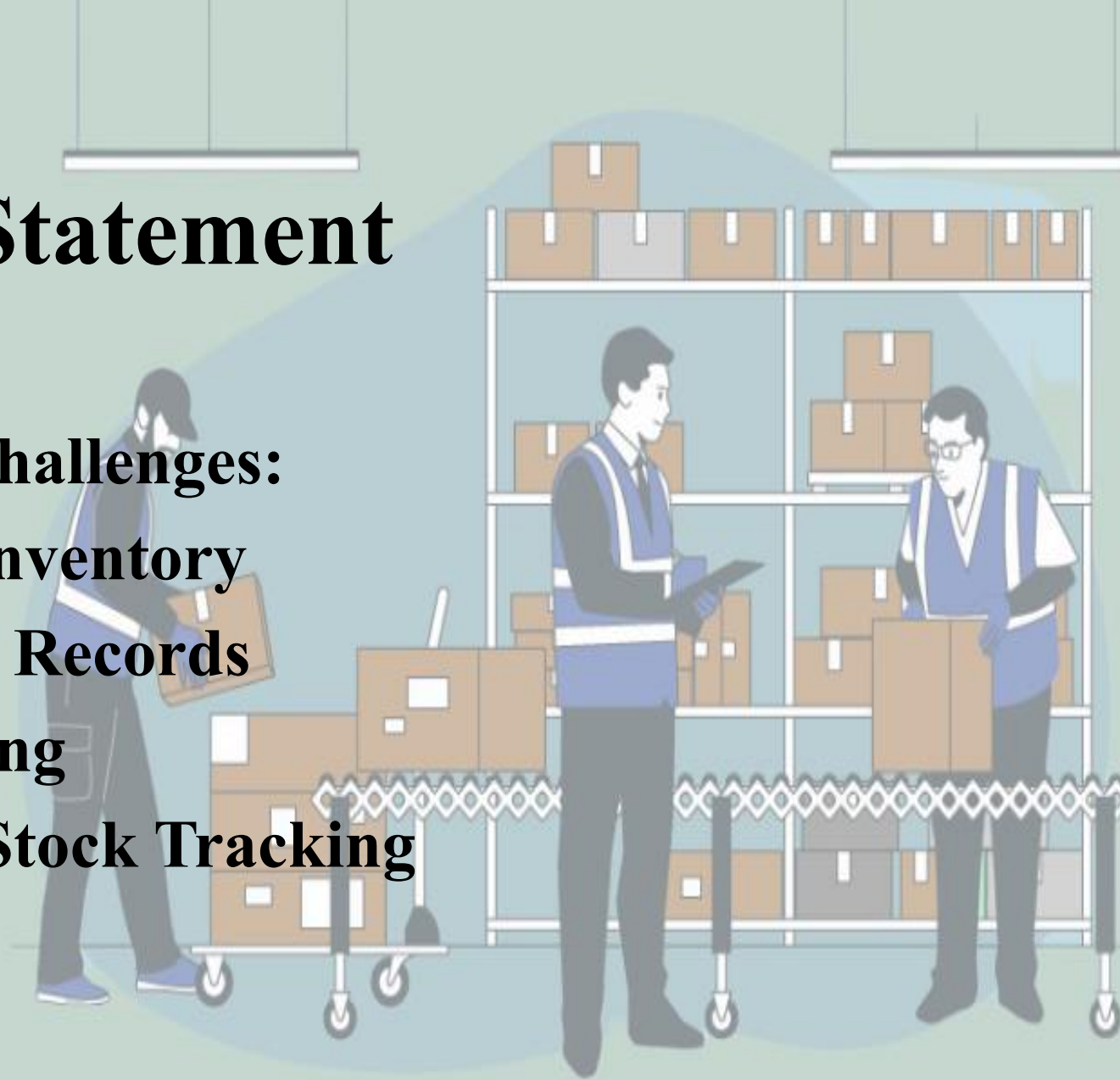
Efficient business operations through a centralized database.



Problem Statement

→ **Current Challenges:**

- ✓ **Manual Inventory**
- ✓ **Duplicate Records**
- ✓ **Slow Billing**
- ✓ **Difficult Stock Tracking**



Objectives

=> Design an EER model that can manage

- ✓ Stores
- ✓ Products
- ✓ Customers
- ✓ Employees
- ✓ Suppliers
- ✓ Orders



Why EER?

ER Model:

Basic entities
Relationships

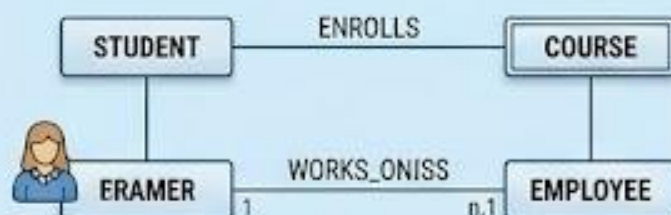


EER Model:

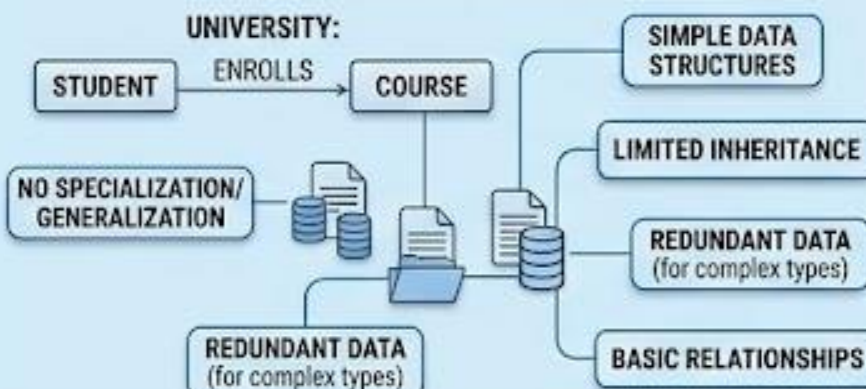
Specialization
Generalization
Better representation

ER vs. EER COMPARISON DIAGRAM

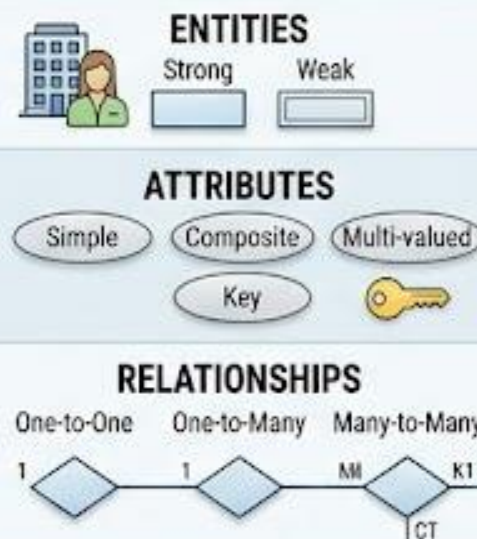
ER MODEL (ENTITY-RELATIONSHIP)



ER MODEL FEATURES



CORE COMPONENTS



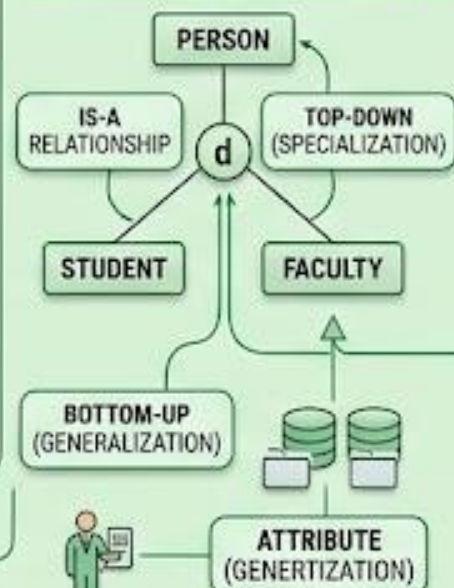
LIMITATIONS

- CHASM TRAP
- FAN TRAP

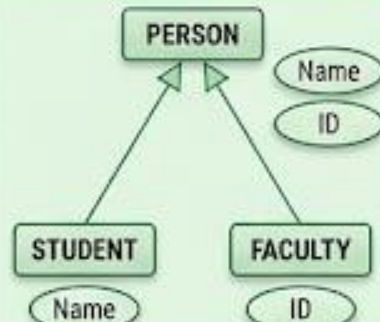
EER MODEL (ENHANCED ENTITY-RELATIONSHIP)

EER MODEL ENHANCEMENTS

SPECIALIZATION/GENERALIZATION



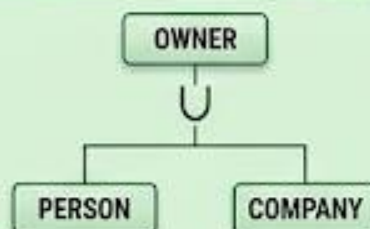
ATTRIBUTE INHERITANCE



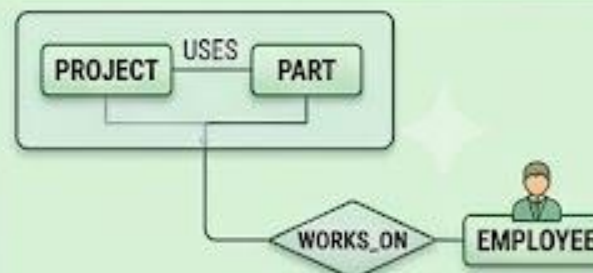
CONSTRAINTS



CATEGORIZATION (UNION TYPE)



AGGREGATION

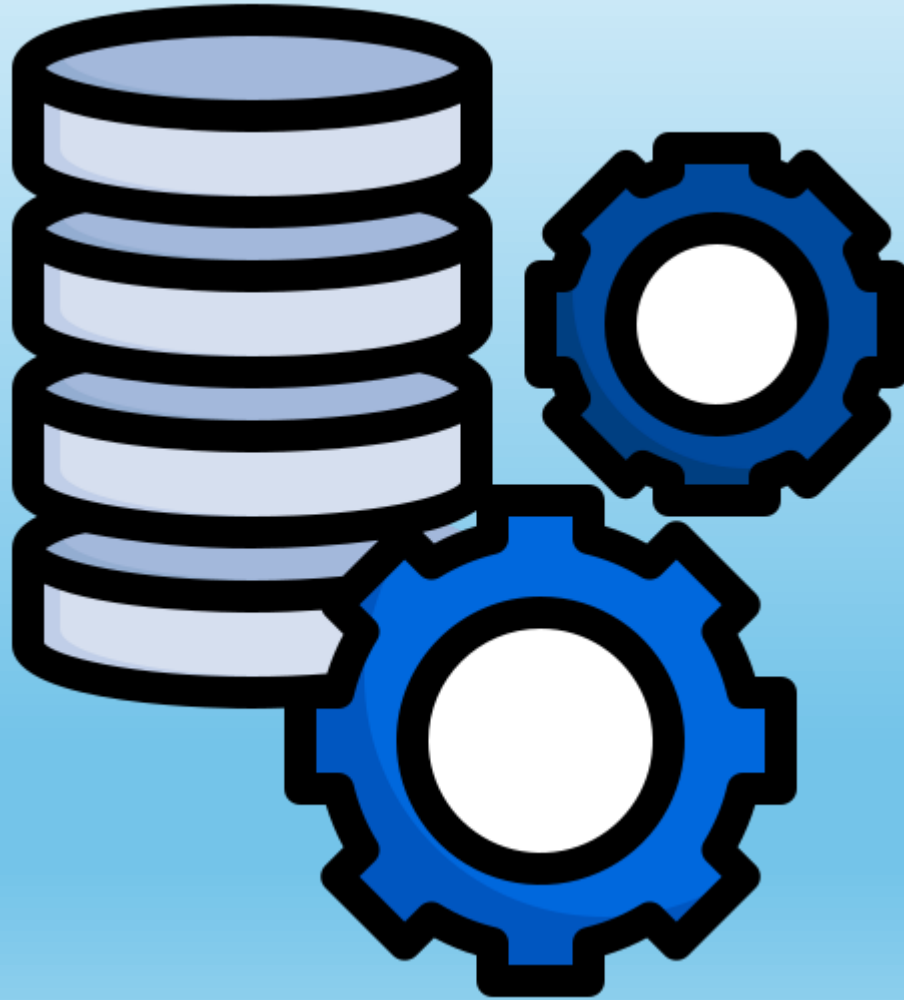


SUMMARY TABLE

FEATURE	ER MODEL	EER MODEL
Hierarchy	Hierarchy verers and hierarchy	Contrasts onn hierarchy
Inheritance	Inheritance	Inheritance attribute inheritance
Complexity	Complexity complexity	Complexity specializatir/generalization
Abstraction	Abstraction data	Abstracted abstraction
Suitable for	Suitable for and contains	Suitable for company or witarement

Main Entities

- Store
- Customer
- Product
- Supplier

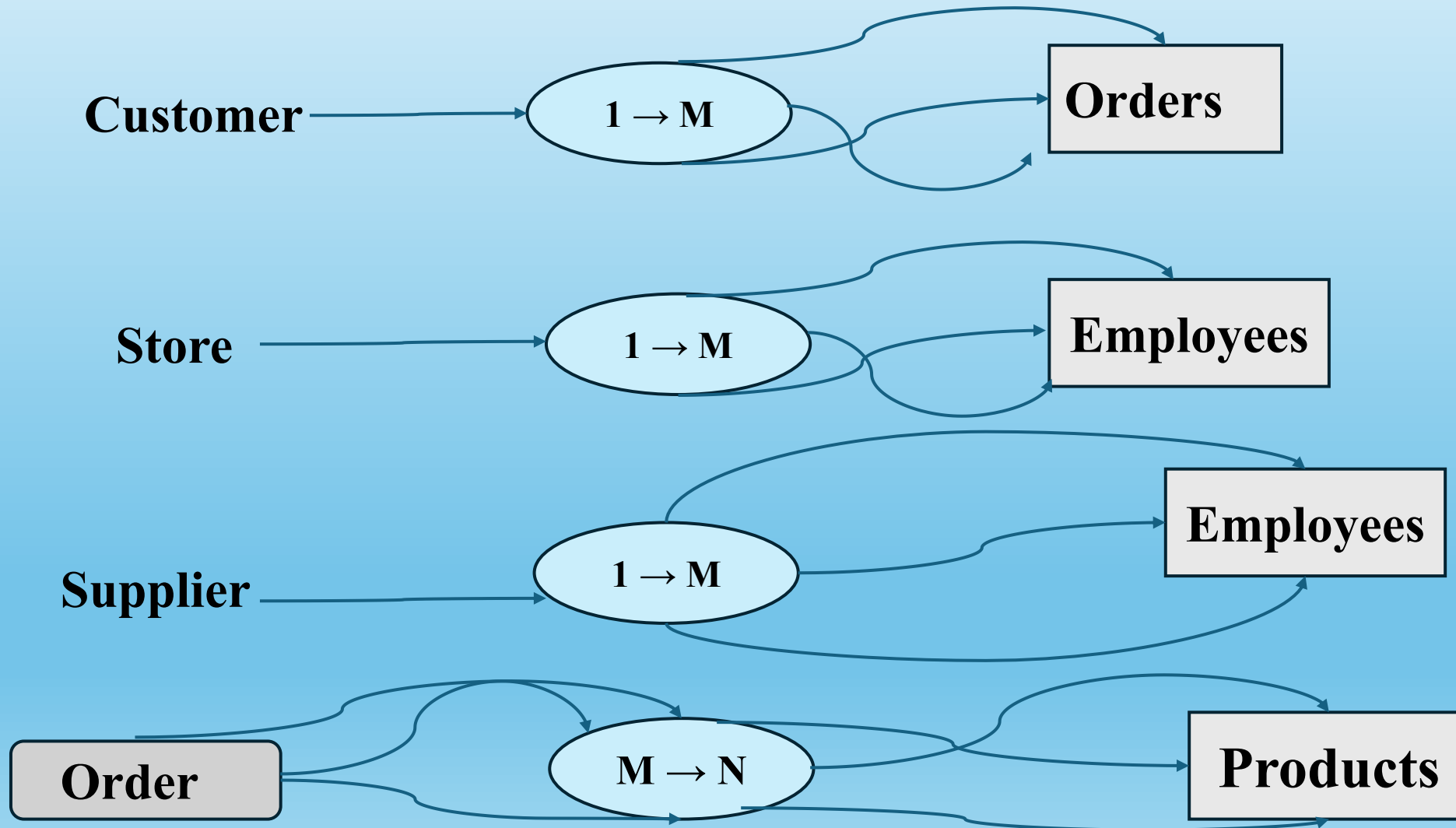


- Order
- Employee
- Inventory
- Payment

Relationships

- Customer → Places → Order
- Store → Contains → Inventory
- Supplier → Supplies → Product
- Employee → Works At → Store
- Order → Contains → Product

Cardinality



Specialization

Employee



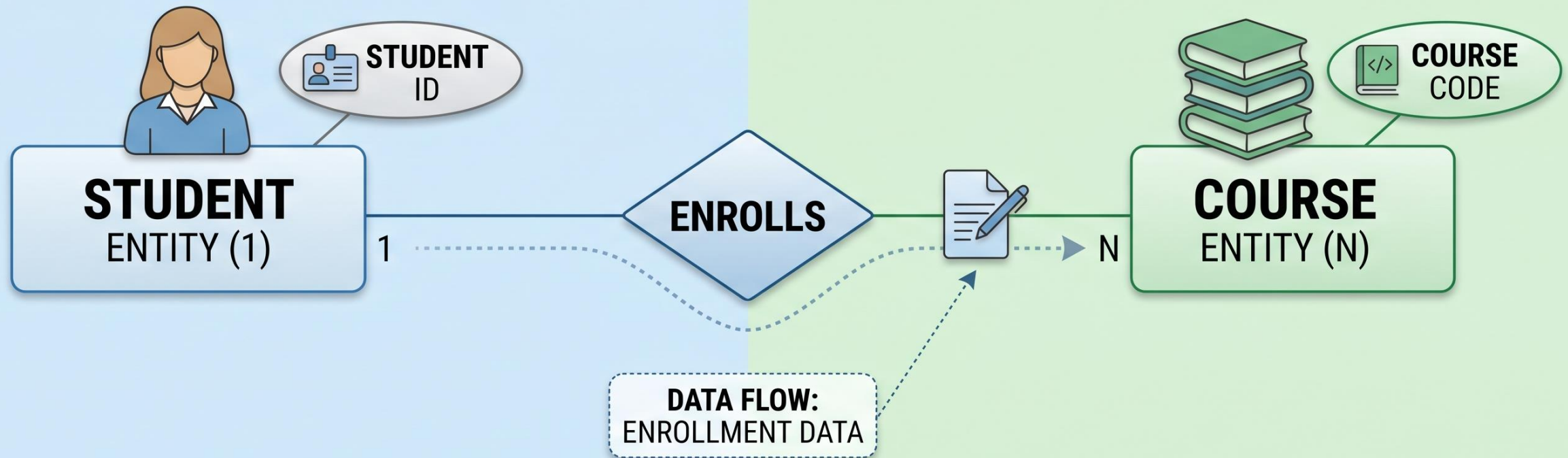
Manager

Cashier

Sales Executive

Warehouse Staff

Simpler ER Relationship & Flow



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THANK

YOU!

